

PES2UG23CS678_VarunNamboodiri

_Lab6

Name: Varun Namboodiri

Section: K

Subject: Cloud Computing

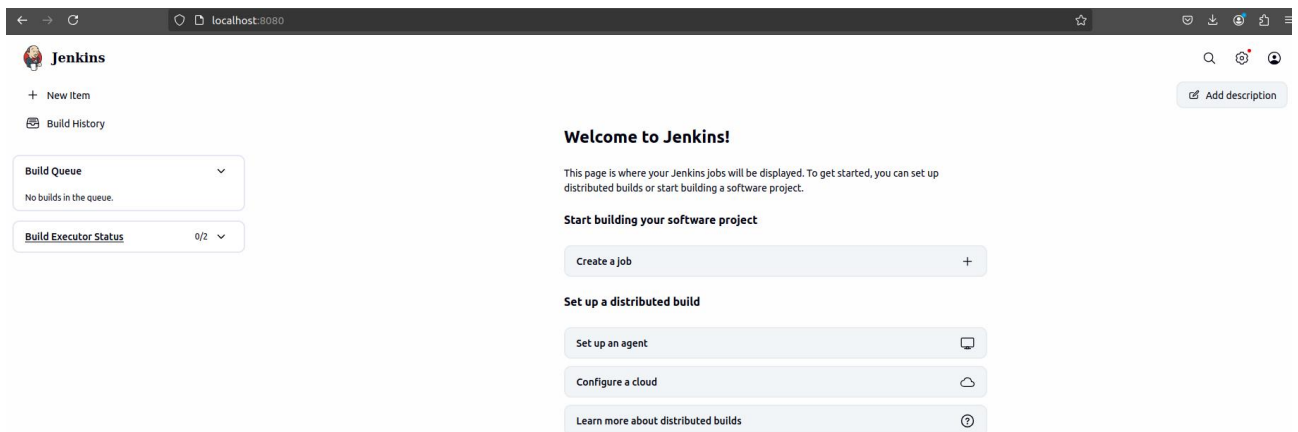
Tasks:-

Task 1:- Set up Jenkins using Docker

Screenshot 1:-

```
pes2ug23cs678@pes2ug23cs678:~/CC_LAB-6$ sudo docker logs jenkins
Running from: /usr/share/jenkins/jenkins.war
webroot: /var/jenkins_home/war
2026-02-17 09:09:59.596+0000 [id=1] INFO winstone.Logger#logInternal: Beginning extraction from war file
2026-02-17 09:10:03.805+0000 [id=1] WARNING o.e.j.ee9.nested.ContextHandler#setContextPath: Empty contextPath
2026-02-17 09:10:04.067+0000 [id=1] INFO org.eclipse.jetty.server.Server#doStart: jetty-12.1.5; built: 2025-12-03T22:18:24.732Z; git: 4905d89d7b6801e792d7b73946cf7b66e4be25d6; jvm 21.0.9+10-LTS
2026-02-17 09:10:05.582+0000 [id=1] INFO o.e.j.e.w.StandardDescriptorProcessor#visitServlet: NO JSP Support for /, did not find org.eclipse.jetty.ee9.jsp.JettyJspServlet
2026-02-17 09:10:05.921+0000 [id=1] INFO o.e.j.s.DefaultSessionIdManager#doStart: Session workerName=node0
2026-02-17 09:10:07.323+0000 [id=1] INFO hudson.WebAppMain#contextInitialized: Jenkins home directory: /var/jenkins_home found at: EnvVars.masterEnvVars.get("JENKINS_HOME")
2026-02-17 09:10:08.294+0000 [id=1] INFO o.e.j.s.handler.ContextHandler#doStart: Started oeje9n.ContextHandler$CoreContextHandler@3681037(Jenkins v2.541.1,/b=file:///var/jenkins_home/war/,a=AVAILABLE,h=oeje9n.ContextHandler$CoreContextHandler$CoreTolstedHandler@2459319c{STARTED})
2026-02-17 09:10:08.444+0000 [id=1] INFO o.e.j.server.AbstractConnector#doStart: Started oejs.ServerConnector@2aa6311a(HTTP/1.1, (http/1.1)){0.0.0.0:8080}
2026-02-17 09:10:08.451+0000 [id=1] INFO org.eclipse.jetty.server.Server#doStart: Started oejs.Server@78525ef9{STARTING}[12.1.5,sto=0] @4388ms
2026-02-17 09:10:08.454+0000 [id=40] INFO winstone.Logger#logInternal: Winstone Servlet Engine running: controlPort=disabled
2026-02-17 09:10:09.069+0000 [id=39] INFO jenkins.model.Jenkins#<init>: Starting version 2.541.1
2026-02-17 09:10:09.493+0000 [id=48] INFO jenkins.InitReactorRunner$1#onAttained: Started initialization
2026-02-17 09:10:09.534+0000 [id=47] INFO jenkins.InitReactorRunner$1#onAttained: Listed all plugins
```

Screenshot 2:-



Task 2:- Create a Jenkins Job to Build the Backend

Screenshot 3:-

```
Jenkins / PES2UG23CS678-backend... / #6

#6 DONE 92.9s

#7 [3/5] WORKDIR /app
#7 DONE 0.2s

#8 [4/5] COPY app.cpp .
#8 DONE 0.1s

#9 [5/5] RUN g++ app.cpp -o app
#9 DONE 2.5s

#10 exporting to image
#10 exporting layers
#10 exporting layers 38.5s done
#10 exporting manifest sha256:1fffe9f48cf75485960fbf7cead429a67cb7132d93f7a838e49548bb922d3ed 0.0s done
#10 exporting config sha256:ac15f90d58bb28c928fa9b432863447c6018356c2403abce5e5448aa81c46dec 0.0s done
#10 exporting attestation manifest sha256:e4bb8cc108e153ac0a7281743c6fb04c0db3efd13bc86514797157b0820db2fd 0.1s done
#10 exporting manifest list sha256:e62e30e49854170af9ab36f884a8123d8f7c5125c3a947cd442b9241025783c4
#10 exporting manifest list sha256:e62e30e49854170af9ab36f884a8123d8f7c5125c3a947cd442b9241025783c4 0.0s done
#10 naming to docker.io/library/backend-app:latest 0.0s done
#10 unpacking to docker.io/library/backend-app:latest
#10 unpacking to docker.io/library/backend-app:latest 11.9s done
#10 DONE 50.7s
Finished: SUCCESS
```

Screenshot 4:-

Jenkins / PES2UG23CS678-backend... 🔍 🔔 👤

Status

</> Changes

Workspace

Build Now

Configure

Delete Project

Git Polling Log

Rename

✓ PES2UG23CS678-backend-build

Permalinks

- Last build (#6), 5 min 17 sec ago
- Last stable build (#6), 5 min 17 sec ago
- Last successful build (#6), 5 min 17 sec ago
- Last failed build (#5), 15 min ago
- Last unsuccessful build (#5), 15 min ago
- Last completed build (#6), 5 min 17 sec ago

Builds >

🔍 Filter /

Today

✓ #6 11:01 AM

✗ #5 10:51 AM

✗ #4 10:45 AM

🔍

🔔

👤

Add description

Task 3:- Parameterized Jenkins job

Screenshot 5:-

Jenkins / PES2UG23CS678-backend... 🔍 🔔 👤

Status

</> Changes

Workspace

Build with Parameters

Configure

Delete Project

Git Polling Log

Rename

Project PES2UG23CS678-backend-build

This build requires parameters:

Backend_Count

1

▶ Build

Cancel

Builds >

🔍 Filter /

Today

✓ #9 11:19 AM

✓ #8 11:19 AM

✗ #7 11:16 AM

✓ #6 11:01 AM

✗ #5 10:51 AM

✗ #4 10:45 AM

✗ #3 10:37 AM

✗ #2 10:34 AM

🔍

🔔

👤

**Jenkins**

PES2UG23CS678-backend...

Status

Changes

Workspace

Build with Parameters

Configure

Delete Project

Git Polling Log

Rename

Project PES2UG23CS678-backend-build

This build requires parameters:

Backend_Count

2

Build

Cancel

Builds >

Filter

Today

#10

2:00 PM

#9

11:19 AM

#8

11:19 AM

#7

11:16 AM

#6

11:01 AM

#5

10:51 AM

#4

10:45 AM

#3

10:37 AM

```
jenkins PES2UG23C5678-backend... / #10

#8 [3/5] WORKDIR /app
#8 CACHED

#9 [5/5] RUN g++ app.cpp -o app
#9 CACHED

#10 exporting to image
#10 exporting layers 0.0s done
#10 exporting manifest sha256:1fffe9f40cf75485960fbf7cead429a67cb7132d93f7a838e49548bb92d3ed done
#10 exporting config sha256:ac15f90d58bb28c928fa9b432863447c6018356c2403abce5e5448aa81c46dec done
#10 exporting attestation manifest sha256:09ec517acf739af0b6680efc7c1168444b995fc8e7f4c7805488c4770f7d09ba 0.1s done
#10 exporting manifest list sha256:eaid38d58a0bb50beec6bf9657b0024378f3f04e73449bc459f48de36c141a0
#10 exporting manifest list sha256:eaid38d58a0bb50beec6bf9657b0024378f3f04e73449bc459f48de36c141a0 0.0s done
#10 naming to docker.io/library/backend-app:latest done
#10 unpacking to docker.io/library/backend-app:latest 0.0s done
#10 DONE 0.3s

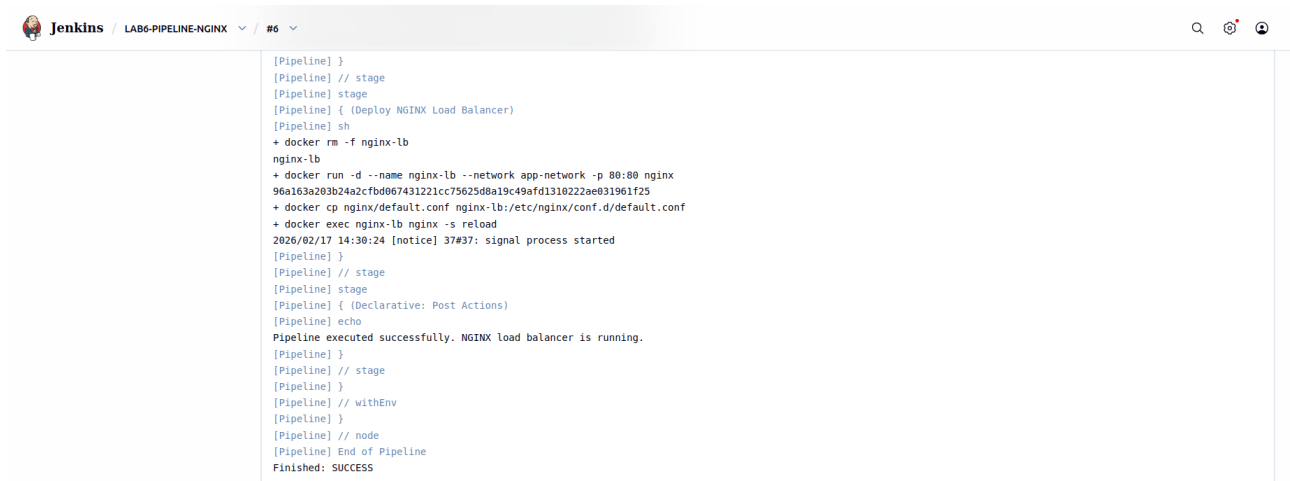
+ docker rm -f backend1 backend2
backend1
backend2
+ [ = 1 ]
+ docker run -d --name backend1 backend-app
9ffe0804d1758674dd53d27999cee6b996c37b7f64e3804bb8b06fd9de49ed80
+ docker run -d --name backend2 backend-app
775d2d3a208e91e26e0fb32d83e3b2338d939502d1ce24d782e5db983bb5e5
Finished: SUCCESS
```

Task 4:- Jenkins Pipeline for Automated Deployment

Screenshot 7:-



Screenshot 8:-



Task 5:- Understanding NGINX Load Balancing Strategies

Screenshot 9:-



Screenshot 10:-



Screenshot 11:-

